

Multivitamins in the Prevention of Cardiovascular Disease in Men: The Physicians' Health Study II Randomized Controlled Trial

Context Though multivitamins aim to prevent vitamin and mineral deficiency, there is a perception that multivitamins may prevent cardiovascular disease (CVD). **Observational studies** examining regular multivitamin use have been inconsistently associated with CVD, with no long-term clinical trials of multivitamin use. **Objective** To determine whether long-term multivitamin supplementation decreases the risk of major cardiovascular events among men. **Design** The Physicians' Health Study II is a randomized, double-blind, placebo-controlled trial of a common daily multivitamin, that began in 1997 with continued treatment and follow-up through June 1, 2011. **Setting and Participants** A total of 14,641 male U.S. physicians initially aged ≥ 50 years (mean [$\pm$ SD] age; 64.3 [$\pm$ 9.2] years), including 754 men with a history of CVD at randomization, were enrolled. **Intervention** Daily multivitamin, as Centrum Silver. **Main Outcome Measures** The primary cardiovascular outcome was a composite endpoint of major cardiovascular events, including nonfatal myocardial infarction (MI), nonfatal stroke, and fatal CVD. Secondary outcomes included MI and stroke individually. **Results** During a median (interquartile range) follow-up of 11.2 (10.7 to 13.3) years, there were 1,732 confirmed major cardiovascular events. Compared with placebo, there was no significant effect of a daily multivitamin on major cardiovascular events (active and placebo multivitamin groups, 11.0 and 10.8 events per 1,000 person-years; hazard ratio [HR], 1.01; 95% confidence interval [CI], 0.91–1.10; $P=0.91$). Further, a daily multivitamin had no effect on total MI (active and placebo multivitamin groups, 3.9 and 4.2 events per 1,000 person-years; HR, 0.93; 95% CI, 0.80–1.09; $P=0.39$), total stroke (active and placebo multivitamin groups, 4.1 and 3.9 events per 1,000 person-years; HR, 1.06; 95% CI, 0.91–1.23; $P=0.48$), or cardiovascular mortality (active and placebo multivitamin groups, 5.0 and 5.1 events per 1,000 person-years; HR, 0.95; 95% CI, 0.83–1.09; $P=0.47$). A daily multivitamin was also not significantly associated with total mortality (HR, 0.94; 95% CI, 0.88–1.02; $P=0.13$). The effect of a daily multivitamin on major cardiovascular events

did not differ between men with or without a baseline history of CVD (P, interaction = 0.62). Conclusions A daily multivitamin did not reduce major cardiovascular events, MI, stroke, and CVD mortality after more than a decade of treatment and follow-up.